Central Limit Theorem and its Application

Chapter 7

Biostatistics and Data Analysis

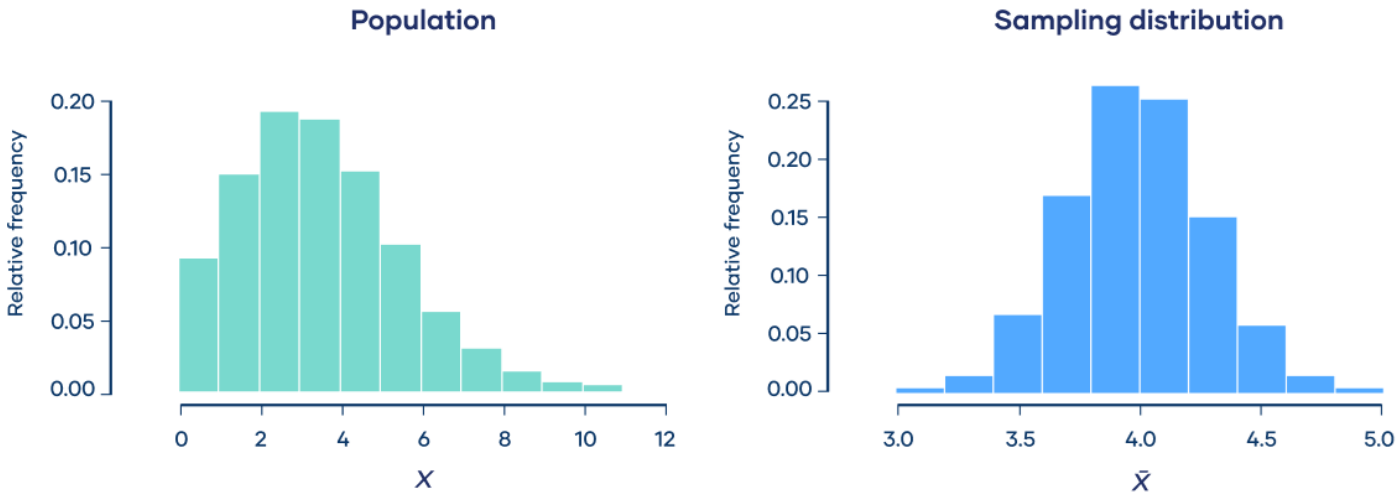
The Central Limit Theorem

- CLT stands for **Central Limit Theorem**.
- It is one of the **most important ideas in statistics**.
- It is the **basis of inferential statistics** (drawing conclusions about a population from samples).
- It explains how the **distribution of sample means** is related to the **population distribution** from which the samples come.

The Central Limit Theorem

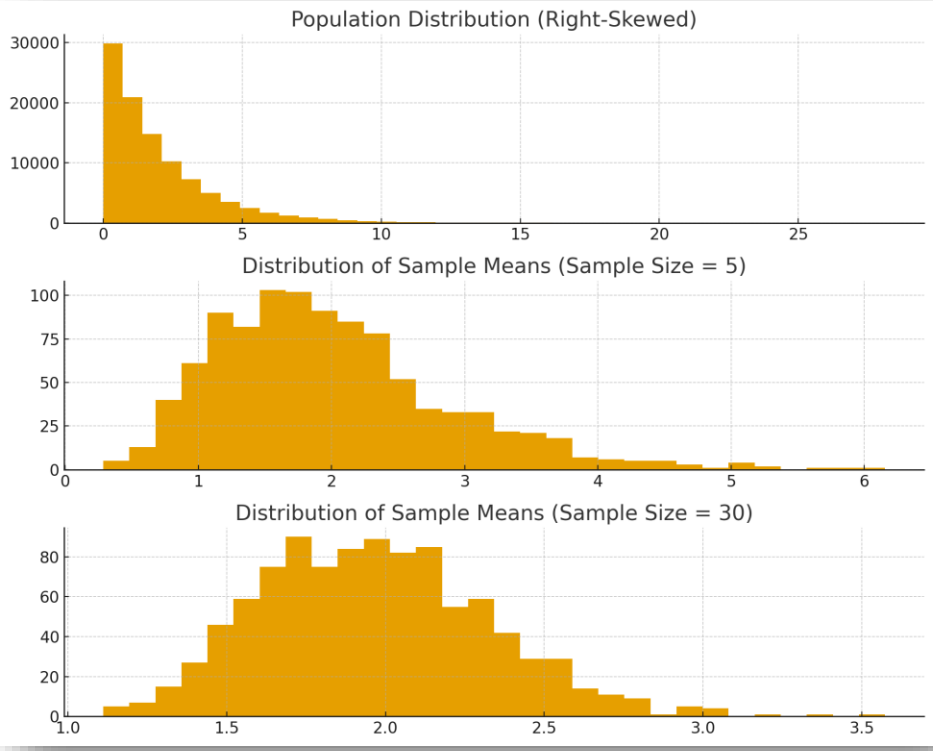
➢ How CLT Works:

- ✓ 1. Take many random samples from a population.
- ✓ 2. Calculate the mean of each sample.
- ✓ 3. Plot the sample means.
- ✓ 4. The resulting distribution will be approximately normal.



The Central Limit Theorem

- The **top graph** shows a **right-skewed population** (not normal).
- The **middle graph** shows sample means for **small samples (n=5)** — still somewhat skewed.
- The **bottom graph** shows sample means for **larger samples (n=30)** — now nearly **normal (bell-shaped)**.



- 👉 This illustrates the **Central Limit Theorem** — as sample size increases, the **distribution of sample means** approaches a **normal distribution**, regardless of the population's shape.

The Central Limit Theorem

What the Central Limit Theorem (CLT) Says:

- When you take **many random samples** from a population:
  - ✓ The **distribution of sample means** becomes **approximately normal**.
  - ✓ The **average of all sample means** equals the **population mean (μ)**.
  - ✓ The **spread (standard deviation)** of sample means equals

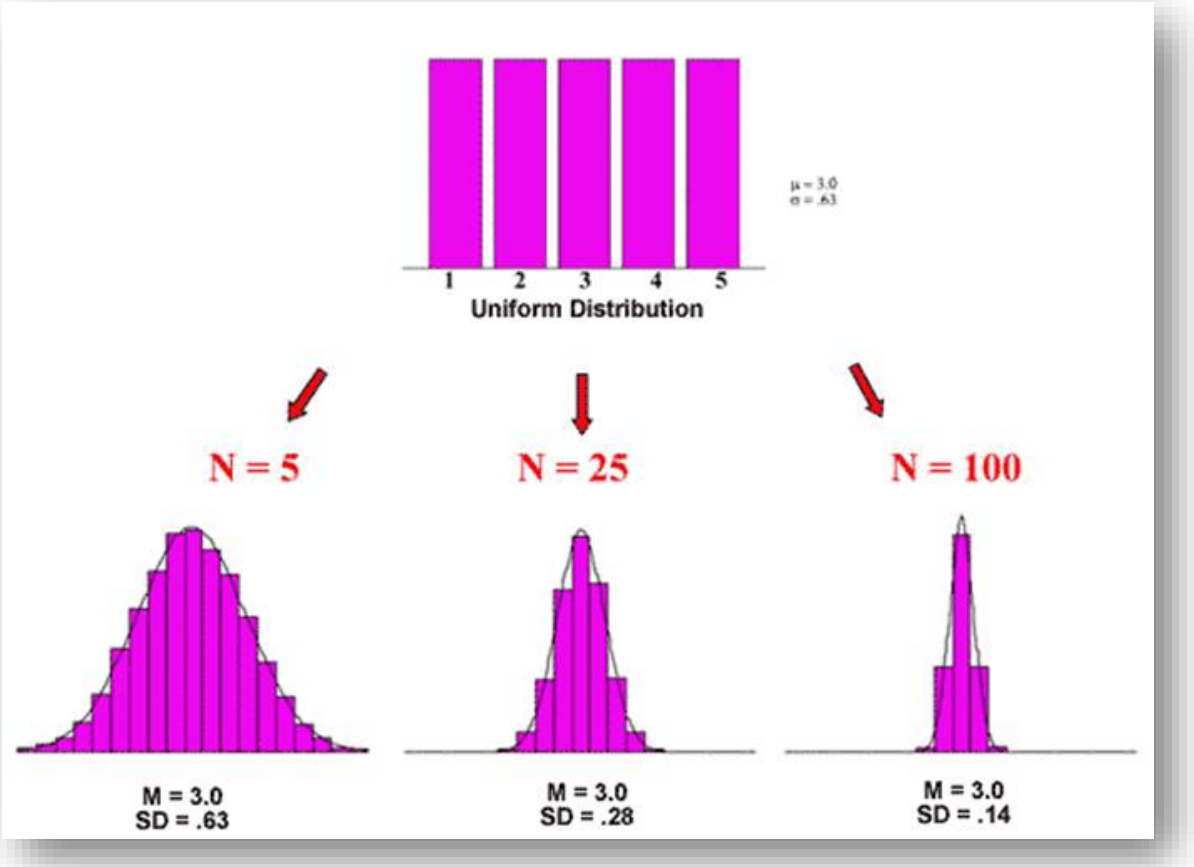
$$\frac{\sigma}{\sqrt{n}}$$

  - where **σ** is the population standard deviation and **n** is the sample size.

The Central Limit Theorem

Key Point:

1. Larger sample size (n) = Less variability.



The Central Limit Theorem

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1. Larger sample size (n) = Less variability.

   - In Simple Words:“ The more data you collect, the closer your sample mean will be to the true population mean, and the less variation you'll see."

- The example regarding students in STA 291
- For a sample of size n=4, the standard error of $\bar{x}$ is

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{0.5}{\sqrt{4}} = 0.25$$

- For a sample of size n=25,

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{0.5}{\sqrt{25}} = 0.1$$

The Central Limit Theorem

If we take **all possible random samples** of size **n** from a population with mean **μ** and standard deviation **σ**, then:

- The **mean of the sample means** is represented as $\mu_{\bar{x}}$, and

$$\mu_{\bar{x}} = \mu$$

  - ✓ This means the average of all sample means equals the population mean.

- The **standard deviation of the sample means** is represented as $\sigma_{\bar{x}}$, and

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

- This value, $\sigma_{\bar{x}}$, is known as the **Standard Error of the Mean (SEM)** — it shows how much the sample means vary from the population mean.

The Central Limit Theorem

- If all possible random samples of size n are selected from a population with mean and standard deviation , the mean of the sample means is denoted by $\mu_{\bar{x}}$, so

$$\mu_{\bar{x}} = \mu$$

- Also the standard deviation of the sample means is denoted by $\sigma_{\bar{x}}$, so

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

- σ_x is called the **standard error of the mean**.

The Central Limit Theorem

CLT applications with examples:

1.Estimating Population Mean

1. *Example:* Use the average height of 100 sampled adults to estimate the city's average height.

2.Hypothesis Testing

1. *Example:* Test if the mean weight of apples from a farm differs from 150g using a sample of 50 apples.

3.Quality Control

1. *Example:* Check if the diameters of bolts in a factory stay within tolerance by sampling 30 bolts.

THANK YOU